

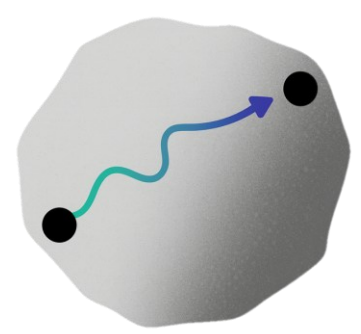
AN EVALUATION OF INVERSION METHODS FOR IN-SITU RF TOMOGRAPHY OF KILOMETRE-SCALE ASTEROIDS

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INTRODUCTION AND MOTIVATIONS

Revealing small-body interiors simultaneously sharpens our grasp of Solar-System formation, resource extraction and bolsters our ability to deflect hazardous objects.

Interior structure of small bodies remain **poorly constrained** despite underpinning planetary defence efforts and Solar System modelling [1]



RF tomography reconstructs a 3D relative-permittivity field from point-to-point radio links *through the body* (e.g., Rosetta/67P) [2]. This implies a structure from *inherent material properties*.

- Wave amplitude attenuates with path length and material variability predominantly due to *absorption* and *multiple scattering* [3]
- Radar uses reflected echoes and different inversion techniques, whereas RF tomography uses constraints along transmitter–receiver paths

This study compares *untuned* arithmetic, statistical, and projected methods while varying asteroid parameters and evaluates them for a lightweight “coin-cell” node concept—a centimetre-scale, gram-level RF unit designed to be deployed as surface networks.

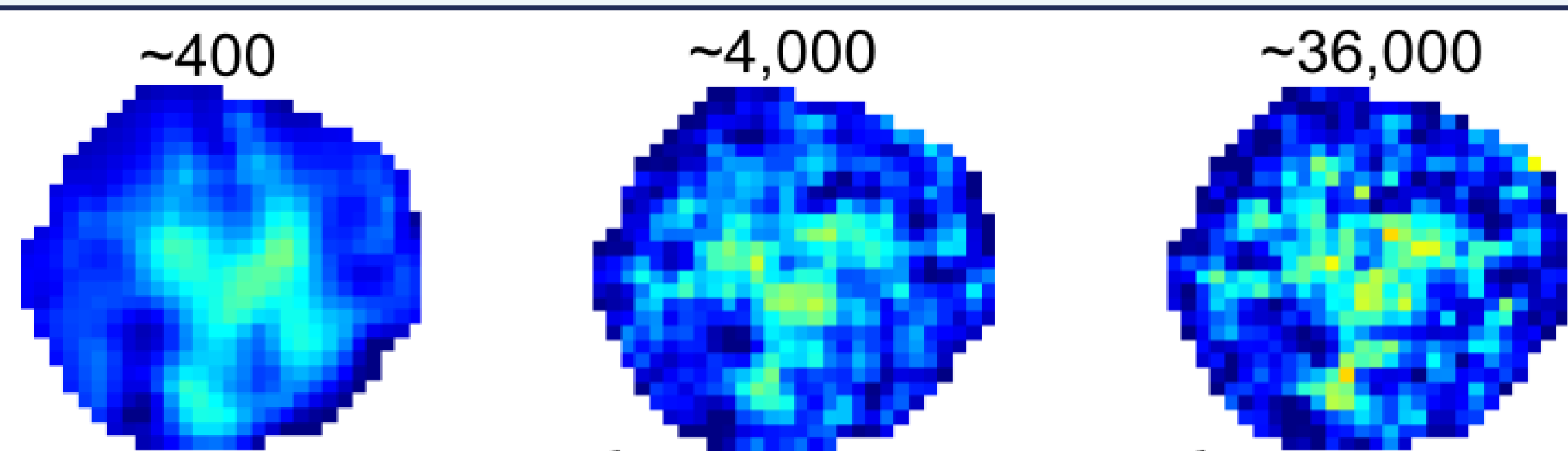


Fig. 1 Bayesian MAP reconstructions of example asteroid at three orders of magnitude of path coverage

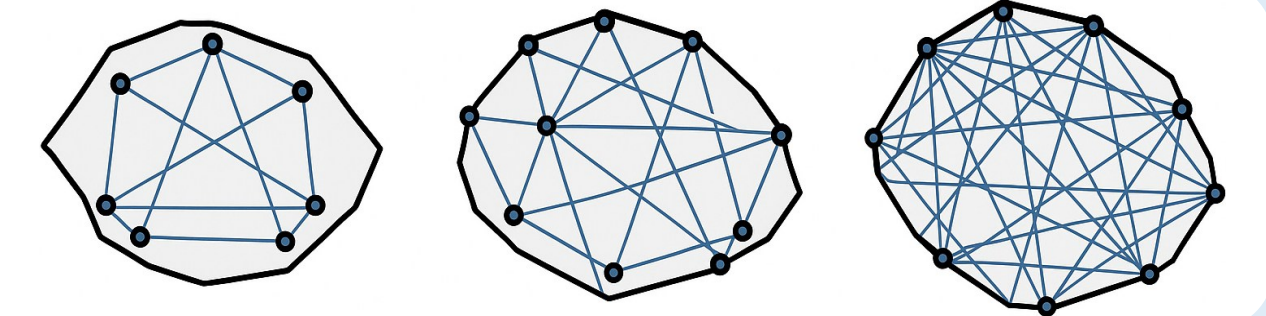
METHODOLOGY



To test which inversion family performs best under realistic mission constraints, we create synthetic RF-tomography datasets across a matrix of asteroid shapes, compositions and path budgets, then benchmark five reconstruction algorithms.

- Shapes:** spinning-top (*Didymos*), elongated (*Oumuamua*) and bilobed (*Itokawa*)
- Geometry:** exact polyhedral meshes voxelated on a common grid and uniformly scaled so total voxel counts match
- Interior compositions:** radial, heterogeneous and mixed (see Fig. 3)
- Unique Paths:** varied by changing the number of surface nodes (example Fig. 2); all unique node pairs used in each case

Fig. 2 Representative unique paths as node numbers are increased on the surface.



Model Generation

- Data:** through-body amplitudes generated with a single-scatter attenuation model (no reflections)
- Reconstructions:** untuned arithmetic, statistical and projected methods — SBP, FBP, ART, SIRT, and BMAP — run in identical environments
- Evaluation:** voxel-wise MAE and RMSE versus the reference volumes, summarised as error–path-count curves

Analysis

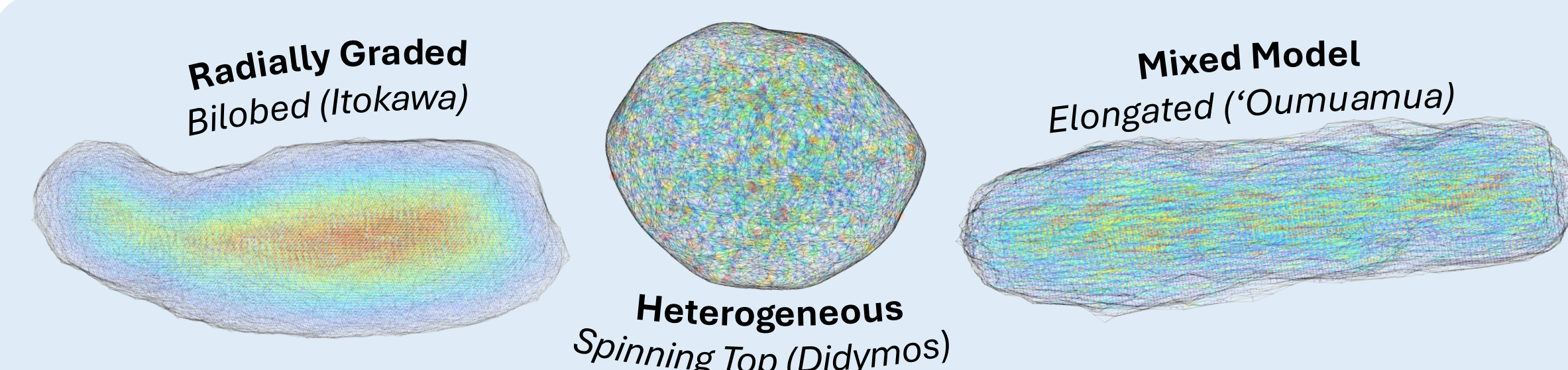
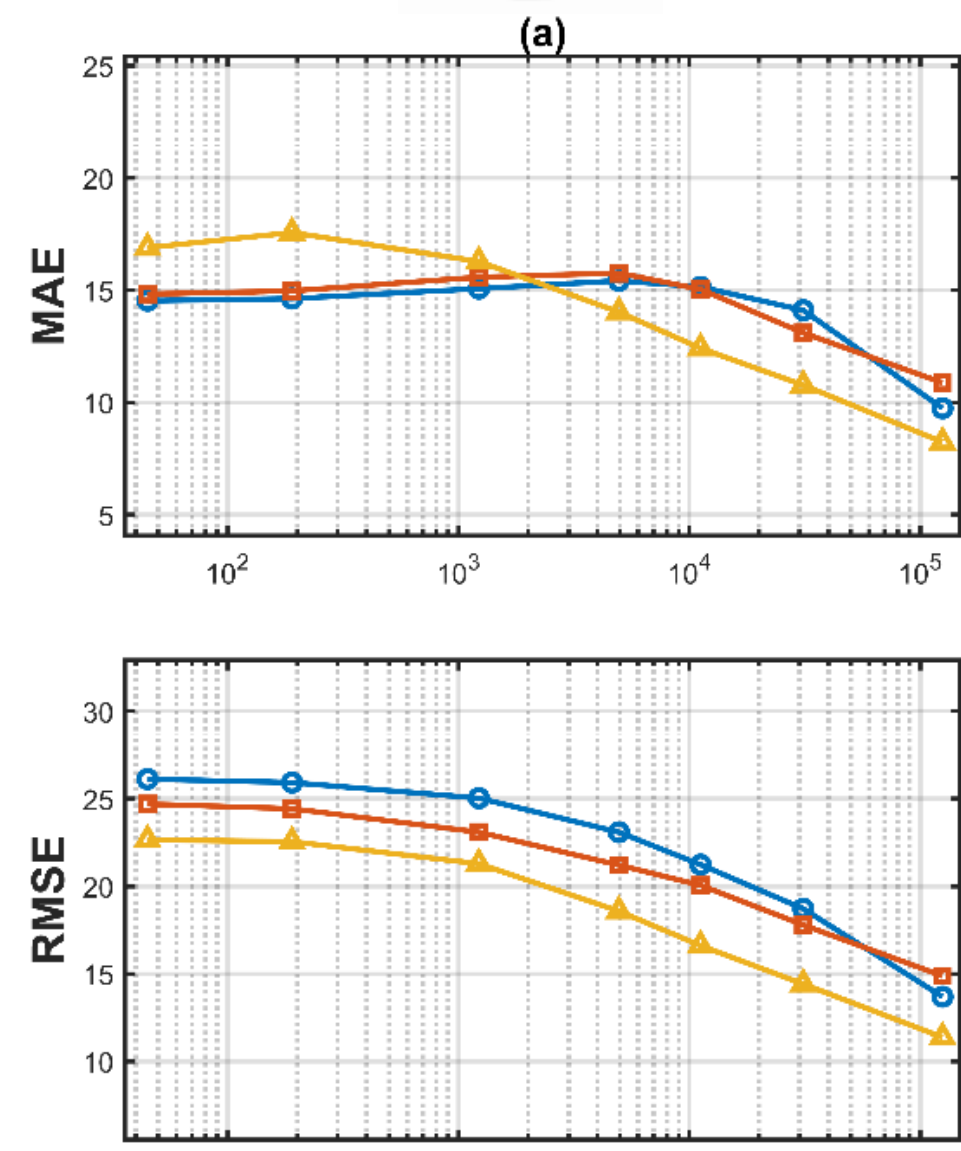


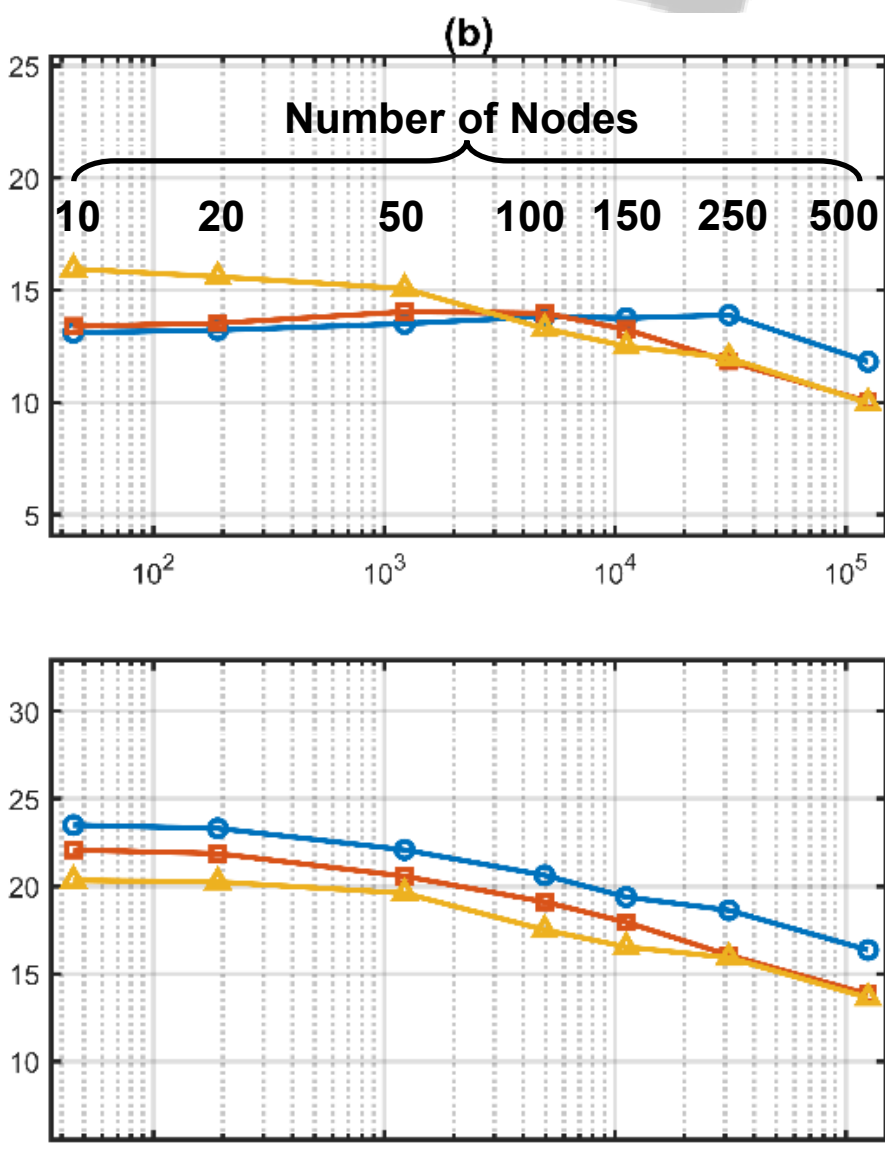
Fig. 3 An example of experimental compositions (**bold**) and shape model (*italic*)

RESULTS AND DISCUSSION

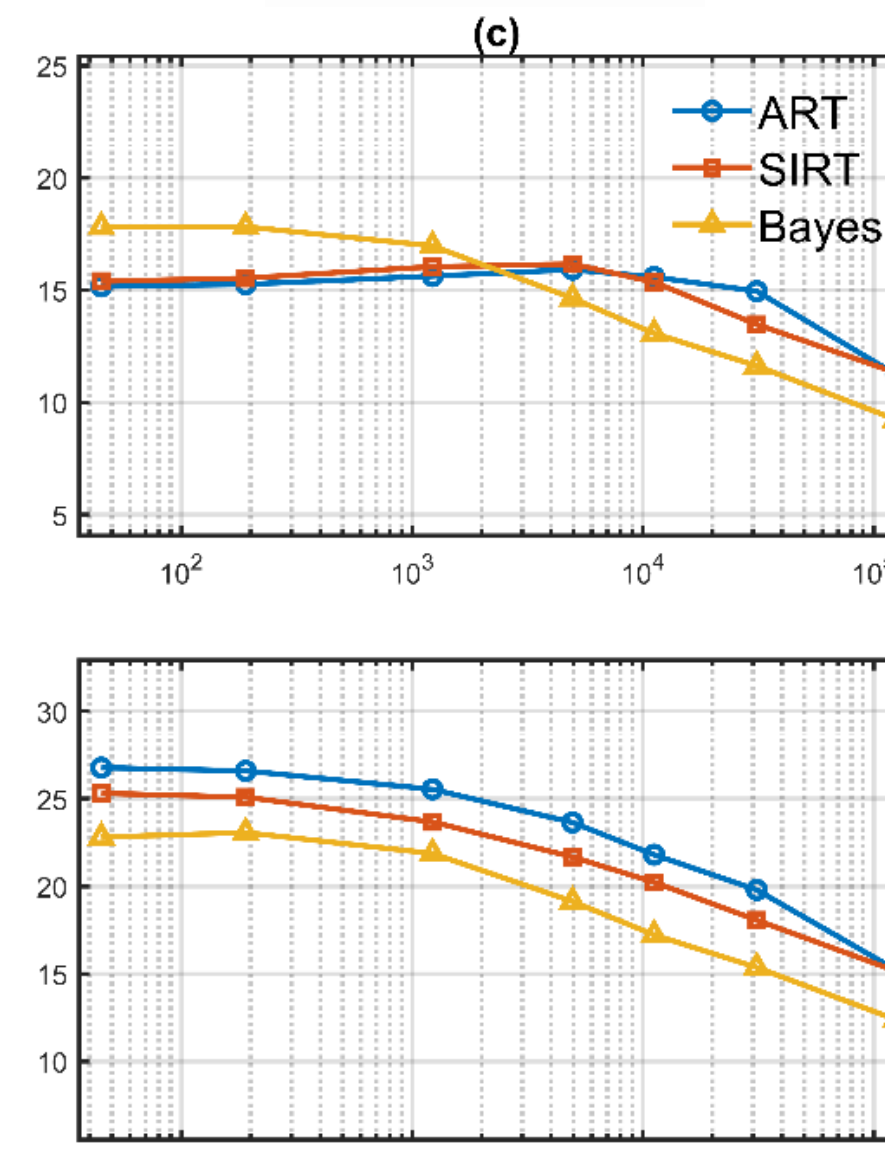
Spinning Top (*Didymos*)



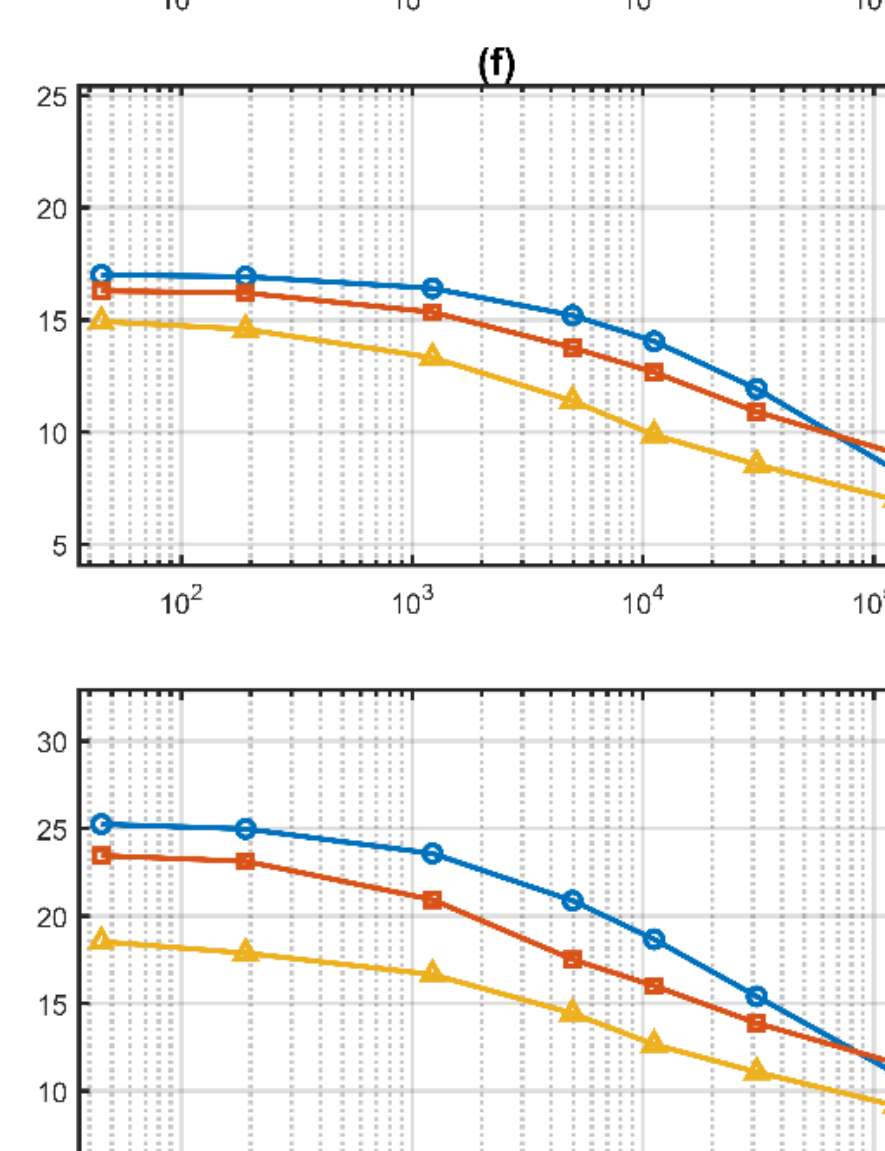
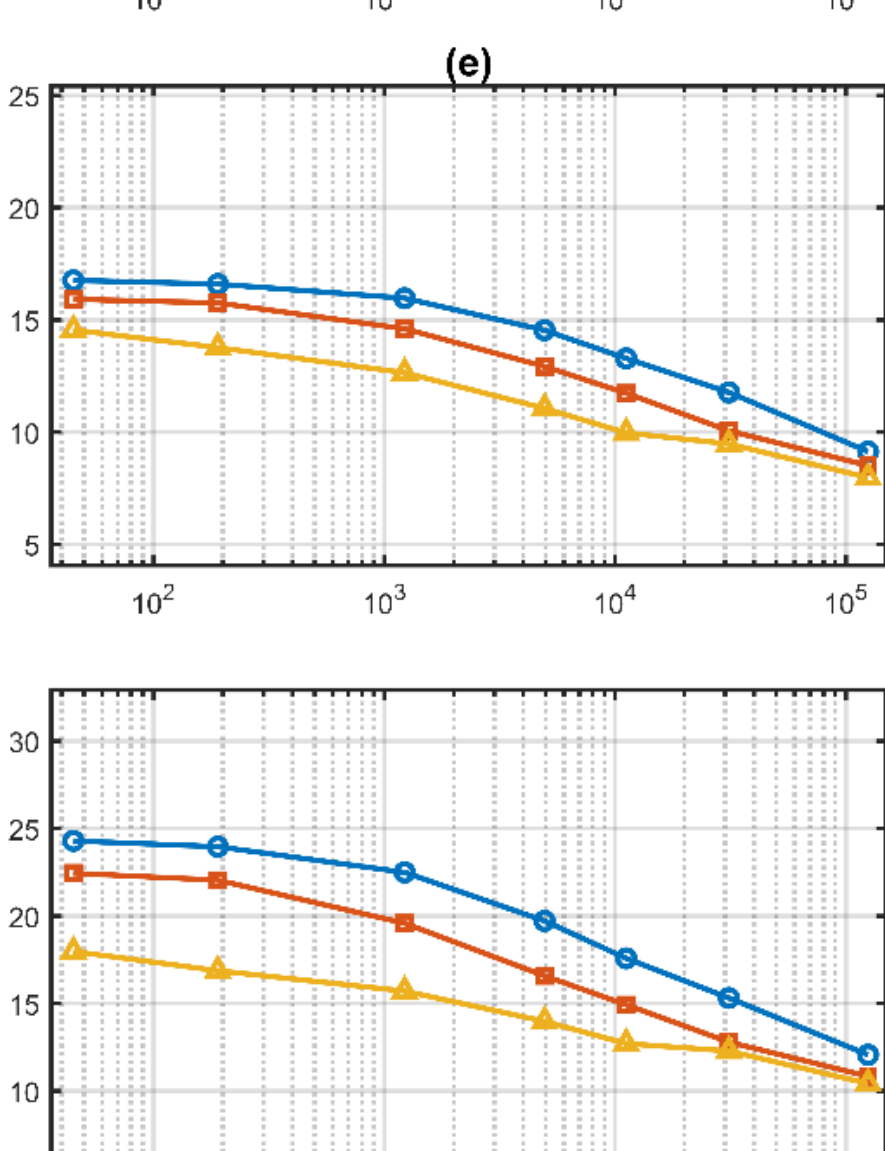
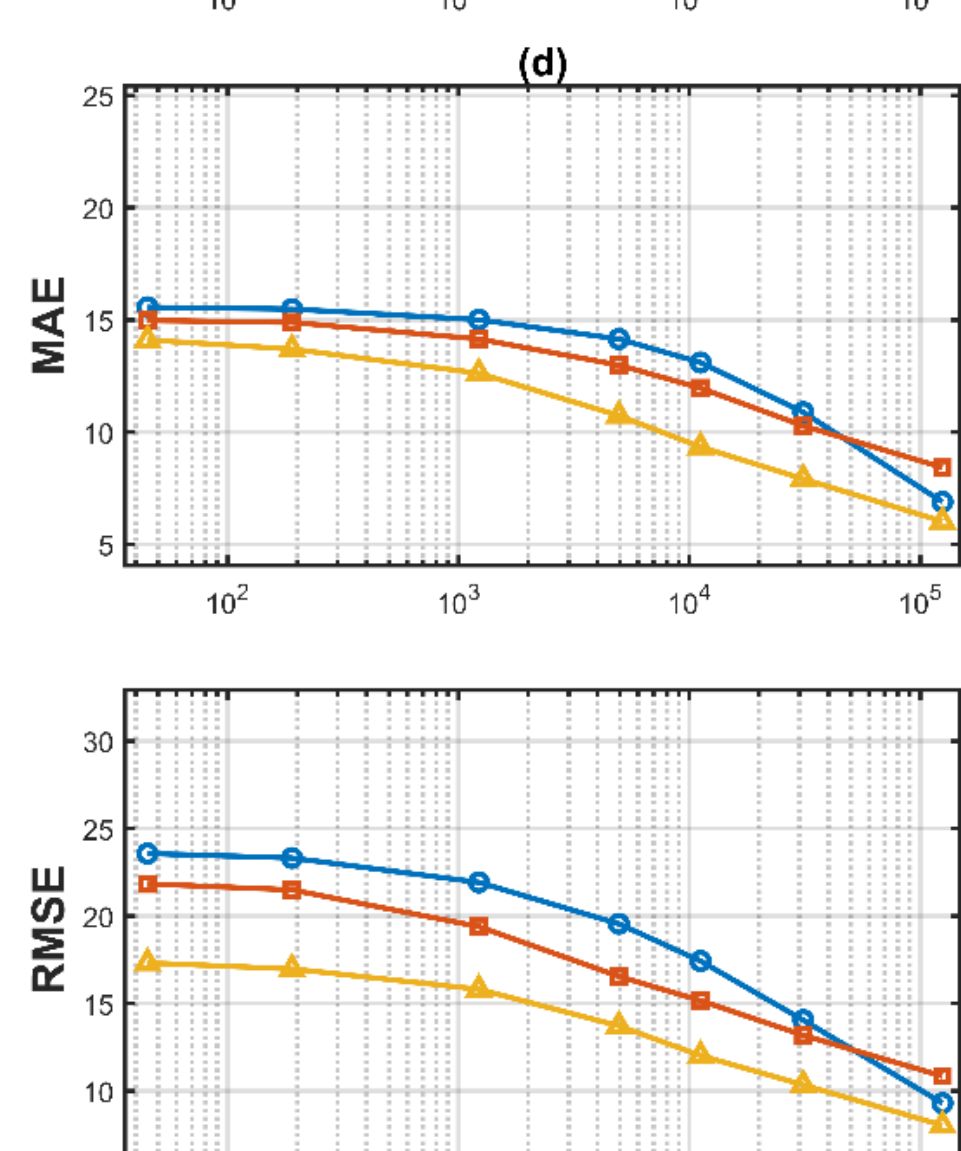
Elongated (*Oumuamua*)



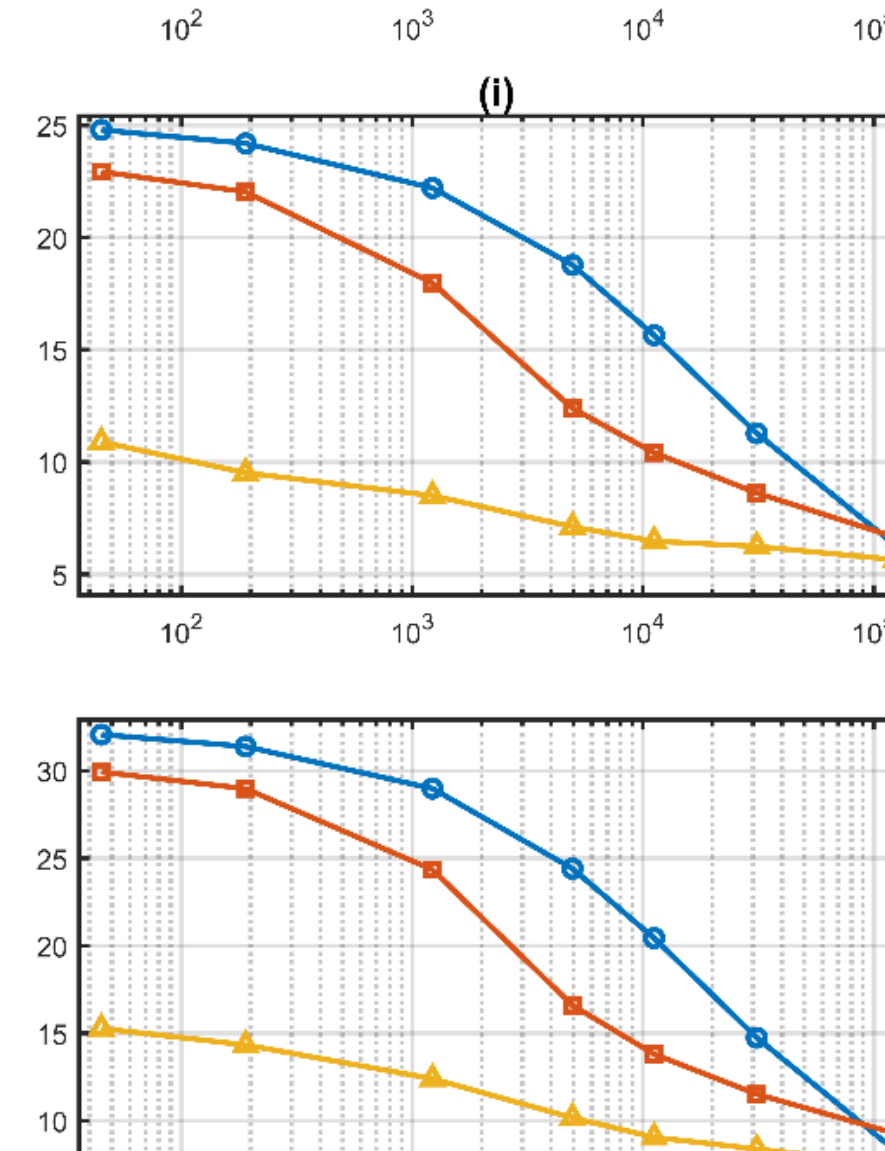
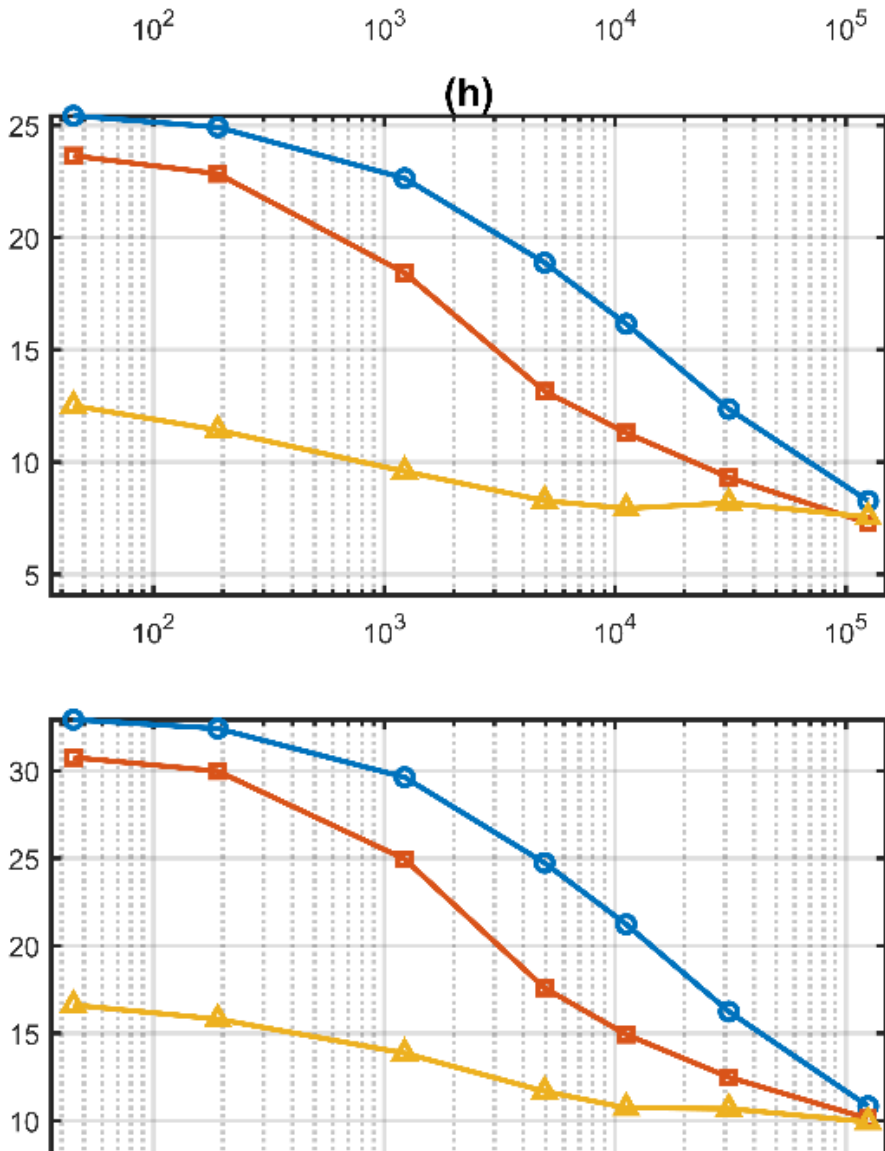
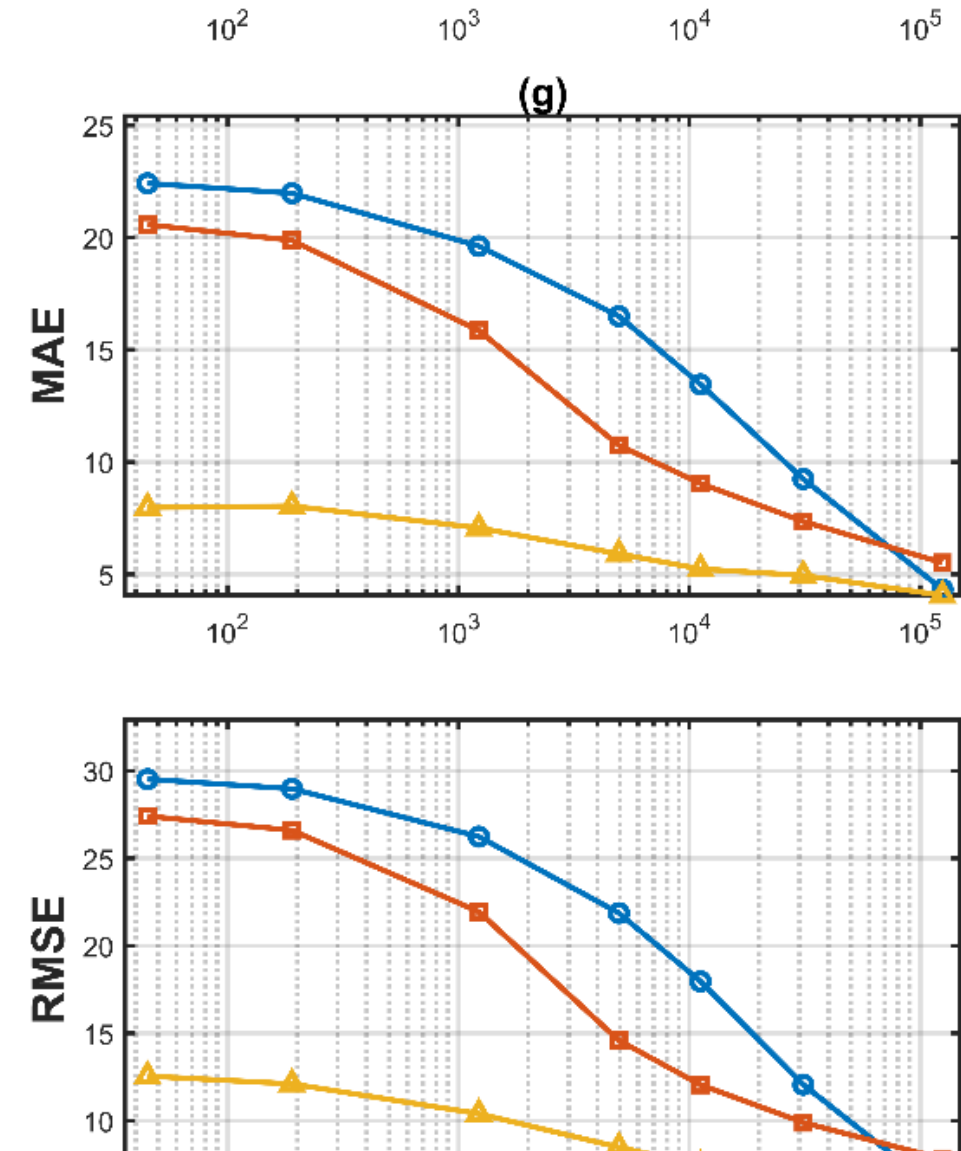
Bilobed (*Itokawa*)



Heterogeneous



Mixed Model



Radially Graded

Number of Unique Paths

Fig. 4 MAE & RMSE of reconstructions through variations in asteroid shape and composition with increasing coverage.

RF-tomography accuracy is driven chiefly by an asteroid's interior composition; algorithm choice and path coverage govern how much of that structure can be recovered. Once coverage is dense, several methods converge, however, in sparse regimes their differences are stark.



Composition: Internal make-up has the largest effect on error; its influence exceeds that of exterior shape.

1

Shape: Elongated and bilobed behave **similarly**; the spinning-top yields the **cleanest** reconstructions, consistent with being closest to a **spherical** model

Arithmetic and Statistical Methods: Bayesian MAP remains the most dependable overall. Even in the most heterogeneous, under-sampled cases it preserves meaningful interior structure—see Fig. 1—despite a higher MAE, and still posts the lowest RMSE by suppressing extremes. ART and SIRT behave similarly: SIRT edges ahead when paths are few, but ART pulls in front once coverage increases

3

Projection algorithms (SBP, FBP):

Demonstrated in Fig. 5, these simple back-projection techniques perform so poorly (greater than twice the MAE of the next worst method in some instances) that including them in the main results would compress the error scale for every other curve.

4

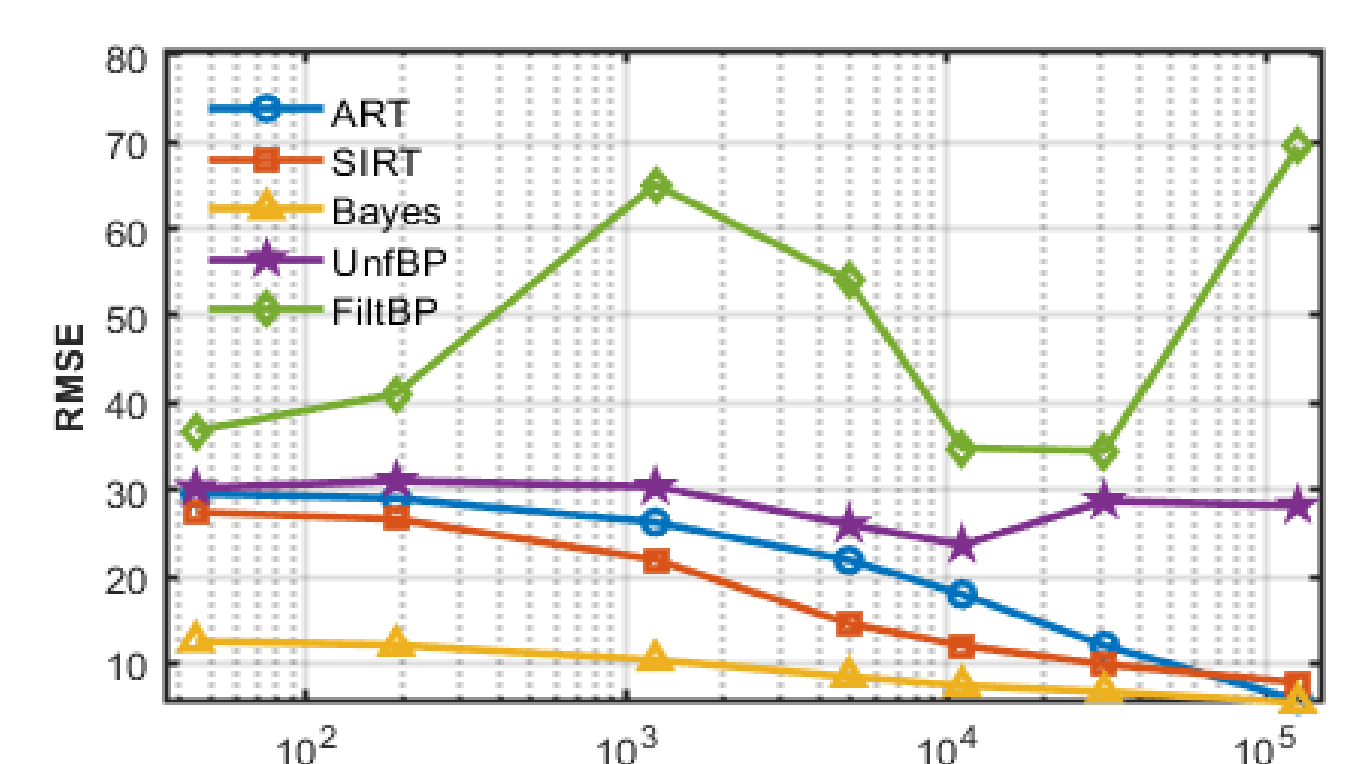


Fig. 5 RMSE for Spinning top-Radially graded configuration, with included projection methods

Path density: Radially graded interiors reconstruct worst with arithmetic methods at low coverage but improve steeply and become the strongest arithmetic results once node pairs are plentiful

Next steps: further shape and composition levels, introduce complementary error metrics (SSIM, MS-SSIM), and benchmark against ML-based approaches, including an ART reconstruction seeded by a BMAP prior.

